

Financial Time Series

TA Session #7

May 7, 2021

Problem 3

$\nwarrow WN(0, \sigma_v^2)$

$$M_{t+1} = M_t + V_t$$

$\nwarrow WN(0, \sigma_w^2)$

$$Y_t = M_t + W_t$$

State Space
model

Goal - to show equivalent to a "Standard Time Series model" in this case ARIMA(0,1,1)

$$\Leftrightarrow \Delta Y_t \sim MA(1)$$

(a) Compute autocorrelation, ACF of ΔY_t

①

$$\Delta Y_t = Y_t - Y_{t-1} = \frac{Y_t - Y_{t-1}}{1} + w_t - w_{t-1}$$

$$\Delta Y_t = w_t - w_{t-1} + v_{t-1}$$

compute autocovariance

$$\gamma(0) = \text{Var}(\Delta Y_t) = \sigma_w^2 + \sigma_w^2 + \sigma_v^2 = 2\sigma_w^2 + \sigma_v^2$$

$$\gamma(1) = \text{Cov}[w_t - w_{t-1} + v_{t-1}, w_{t-1} - w_{t-2} + v_{t-2}]$$

$$= \text{Cov}[-w_{t-1}, w_{t-1}] = -\sigma_w^2$$

$$\gamma(0), \gamma(1), \gamma(k) = 0 \quad k > 1$$

6) equate with general ~~MA~~ $MA(1)$

$$\begin{aligned} \gamma(0), \gamma(1) & \quad X_t = \varepsilon_t + \theta \varepsilon_{t-1} \\ \text{functions of } \theta, \sigma & \quad \{\varepsilon_t\} \sim WN(0, \sigma^2) \end{aligned}$$

(2)

equate autocovariance of δ_y with δ_x
ie find θ, σ as a function of

σ_w^2, σ_v^2 solve quadratic eqn
two ~~two~~ roots

$$(c) \sigma_w^2 = 8, \sigma_v^2 = 20 \Leftrightarrow (\theta_1, \sigma_1), (\theta_2, \sigma_2)$$

(d) simulate both $(\theta_1, \sigma_1), (\theta_2, \sigma_2)$
see if they differ.